

EIS Motion System – Operating Guide

Hardware: BTT SKR Mini E3 v2.0 with TMC2209 drivers. Marlin firmware. Up to 4 stepper motors (X, Y, Z, E).

1. Initial Setup

1. Install Pronterface from <https://www.pronterface.com> (download Printrun, run pronterface.exe).
2. Power on the board (24V supply) BEFORE connecting USB.
3. Connect USB cable from board to computer.
4. In Pronterface: select the COM port assigned to the board, set baud rate to 115200, click Connect.
5. Confirm the message log shows "Printer is now online."

2. Pre-Flight Checklist (every session)

- All motor cables firmly seated at both ends (board and motor).
- 24V power supply on, board power LED lit.
- Pronterface connected, "Printer is now online." displayed.
- Send M122. Verify output matches the reference on Page 2 (all drivers report OK).
- Reset coordinates with G92 X0 Y0 Z0 E0 before sending move commands.
- Set relative positioning with G91 (use G90 for absolute).

3. Key G-Code Commands

Command	Purpose	Example
M122	Driver status report. Check before and after moves.	M122
G91	Relative positioning mode (move BY this amount).	G91
G90	Absolute positioning mode (move TO this position).	G90
G92	Reset current position to specified values. Does not move motors.	G92 X0 Y0 Z0 E0
G1	Linear move. F = feedrate in mm/min. Combine axes for simultaneous motion.	G1 X10 Y10 F600
M906	Set/report motor current in mA RMS.	M906 X800 Y800
M500 / M501 / M502	Save / load / reset to firmware defaults (EEPROM).	M500
M503	Report all current settings (steps/mm, feedrates, currents, etc.).	M503
M84	Disable all stepper motors (release holding current).	M84

4. Software Travel Limit (Important)

Marlin enforces software bounds: X/Y are capped at 235mm, Z at 250mm. Once cumulative position reaches the cap, further moves in that direction are silently ignored — no error, no warning, motors just stop responding.

Workaround: reset coordinates whenever you approach the limit, or before each batch of test moves:

```
G91                ; relative positioning
G92 X0 Y0 Z0 E0    ; reset position to 0
G1 X10 Y10 Z10 F600 ; safe move
```

If a motor stops responding mid-session, run G92 X0 Y0 Z0 E0 first before assuming hardware fault.

5. Typical Workflow

```
M122              ; verify drivers OK
G91               ; relative mode
G92 X0 Y0 Z0 E0    ; reset coordinates
G1 X10 F600        ; move X 10mm at 10 mm/s
G1 X10 Y10 Z10 E10 F600 ; move all four axes simultaneously
G1 X-10 Y-10 Z-10 E-10 F600 ; reverse
M84               ; release motors when done
```

Troubleshooting & Reference

M122 Reference – Healthy Output

This is what M122 should look like at idle when everything is working. With no motor connected to an axis, ola/olb (open load) flags will show — that is expected and not an error.

```
>>> M122
      X      Y      Z      E
Set current 800    800    800    800
RMS current 795    795    795    795
vsense              1=.18 1=.18 1=.18 1=.18
stealthChop false  false  false  false
msteps              16     16     16     16
interp              true   true   true   true
OT prewarn  false  false  false  false
DRVSTATUS  X      Y      Z      E
olb         *      *      *      *    <-- expected if no motor connected
ola         *      *      *      *    <-- expected if no motor connected
Driver registers:
      X      0x80:0C:00:C0
      Y      0x80:0C:00:C0
      Z      0x80:0C:00:C0
      E      0x80:0C:00:C0
Testing X connection... OK
Testing Y connection... OK
Testing Z connection... OK
Testing E connection... OK
```

Error Signatures

Symptom in M122	Likely Cause	Action
Bad response! / Error: All LOW / TMC CONNECTION ERROR notification	Driver UART crashed under load. Almost always a motor cable issue (miswired or intermittent short).	Power cycle the board. Verify motor cable wiring (see section below). Do not retry moves until cable is verified.
ola/olb flags on a connected motor	Open load. Motor not electrically connected through cable, or coil is broken.	Reseat cable. Multimeter test motor coils (see section below).
s2ga / s2gb / s2vsa / s2vsb flags	Short to ground or supply on motor coil. Cable damage or coil-pair miswire.	Power off immediately. Inspect cable. Verify coil pair routing matches board pinout.
ot or otpw flags	Driver overtemperature. Current too high or no airflow.	Lower current with M906. Verify board fan running.
Settings reverted after power cycle	EEPROM not saved or firmware default override.	Send M502, then M500, then power cycle. Verify with M122.
Motor stops responding after	Hit software endstop limit (X/Y	Send G92 X0 Y0 Z0 E0 to

Symptom in M122	Likely Cause	Action
several moves	> 235mm, Z > 250mm).	reset position, then resume.

Verifying Motor Coil Wiring

Most motor failures on this board trace to incorrect cable wiring. Pre-made cables are not universally compatible — different motors use different pin orderings. Verify before powering up.

Step 1: Identify motor coil pairs

With the motor disconnected from the cable, use a multimeter on the lowest resistance range. A 4-wire bipolar stepper has two coils; each coil pair will read 1–4 Ω between its two wires, and infinite/open between wires of different coils.

Compare your measurements to the motor's datasheet pin assignments. The two pins that show low resistance to each other are one coil. The other two pins are the second coil.

Step 2: Match coil pairs to board pinout

The SKR Mini E3 v2.0 motor connectors are labeled (from left to right): 2B, 1B, 1A, 2A. The driver coil pairs are:

- Coil 1 = pins 1A and 1B (the two MIDDLE pins)
- Coil 2 = pins 2A and 2B (the two OUTER pins)

Each motor coil's two wires must terminate on the same driver coil pair. Splitting a motor coil across both driver pairs (e.g., one wire to 1A and the other to 2B) will cause UART disconnect on first move.

Step 3: Verify cable continuity end-to-end

With cable disconnected from both motor and board, multimeter from each pin on the motor-side connector to each pin on the board-side connector. Document the mapping. Then verify:

- Both wires of motor coil 1 land on board pins 1A and 1B (in either order).
- Both wires of motor coil 2 land on board pins 2A and 2B (in either order).

If wrong, depin and re-arrange the connector pins, or use a different cable.

Safety Notes

- NEVER unplug a motor while the board is powered. The driver MOSFETs can be damaged by inductive flyback.
- If TMC CONNECTION ERROR appears, power cycle before further commands. Repeated motion attempts in fault state can permanently damage the driver.
- Do not exceed ~1400 mA RMS without verifying motor rating and adding heatsinks/airflow on the drivers.
- Send M84 to release motors when done — prevents unnecessary heating from holding current.

Calibration: Settings to Update for Final Setup

All settings below live in the Marlin firmware source. Changes require recompiling and re-flashing the board. After flashing, send M502 then M500 to load defaults into EEPROM.

File: Configuration.h

Setting	Current Value	How to Update
DEFAULT_AXIS_STEPS_PER_UNIT (line 815)	{ 200, 200, 200, 200 }	Recalculate per axis: steps/mm = (motor_steps_per_rev × microsteps) /

Setting	Current Value	How to Update
		mm_per_rev. Order is { X, Y, Z, E }.
INVERT_X_DIR / INVERT_Y_DIR / INVERT_Z_DIR (lines 1199–1201)	true / true / false	Flip true/false to reverse motor direction. Do NOT swap motor wires; change firmware instead.
INVERT_E0_DIR (line 1206)	true	Same — flip if E direction is wrong.
X_BED_SIZE / Y_BED_SIZE (lines 1241–1242)	235 / 235	Set to actual physical X and Y travel limits in mm.
Z_MAX_POS (line 1250)	250	Set to actual Z travel limit in mm.
DEFAULT_MAX_FEEDRATE (line 822)	{ 200, 200, 5, 25 }	Max mm/s per axis. Lower if motors skip at high speeds.
DEFAULT_MAX_ACCELERATION (line 835)	{ 500, 500, 100, 5000 }	Max mm/s ² per axis. Reduce if motors stall during accel.
DEFAULT_ACCELERATION (line 850)	100	Default print acceleration in mm/s ² .

File: Configuration_adv.h

Setting	Current Value	How to Update
X_CURRENT / Y_CURRENT / Z_CURRENT (lines 2497/2515/2533)	580 mA each	RMS current in mA. Set to 70–85% of motor's rated current. Can also be changed live with M906.
E0_CURRENT (line 2569)	650 mA	Same — set to extruder motor rating.
X_MICROSTEPS (line 2499, same for Y/Z/E0)	16	Microsteps per full step. If changed, DEFAULT_AXIS_STEPS_PER_UNIT must be recalculated.

Calibration order (when first commissioning a new mechanical setup)

- Set X_BED_SIZE, Y_BED_SIZE, Z_MAX_POS to actual travel limits.
- Verify motor directions by jogging each axis. Flip INVERT_*_DIR if any axis moves wrong direction.
- Calculate DEFAULT_AXIS_STEPS_PER_UNIT for each axis: command a known distance, measure actual travel, scale steps/mm to match.
- Set motor currents to lowest value that doesn't skip steps under expected mechanical load.
- Verify max feedrate and acceleration — reduce if motors stall during fast moves or hard accel.
- After flashing: M502 → M500 → M122 to confirm settings applied.